

Service Oriented Computing

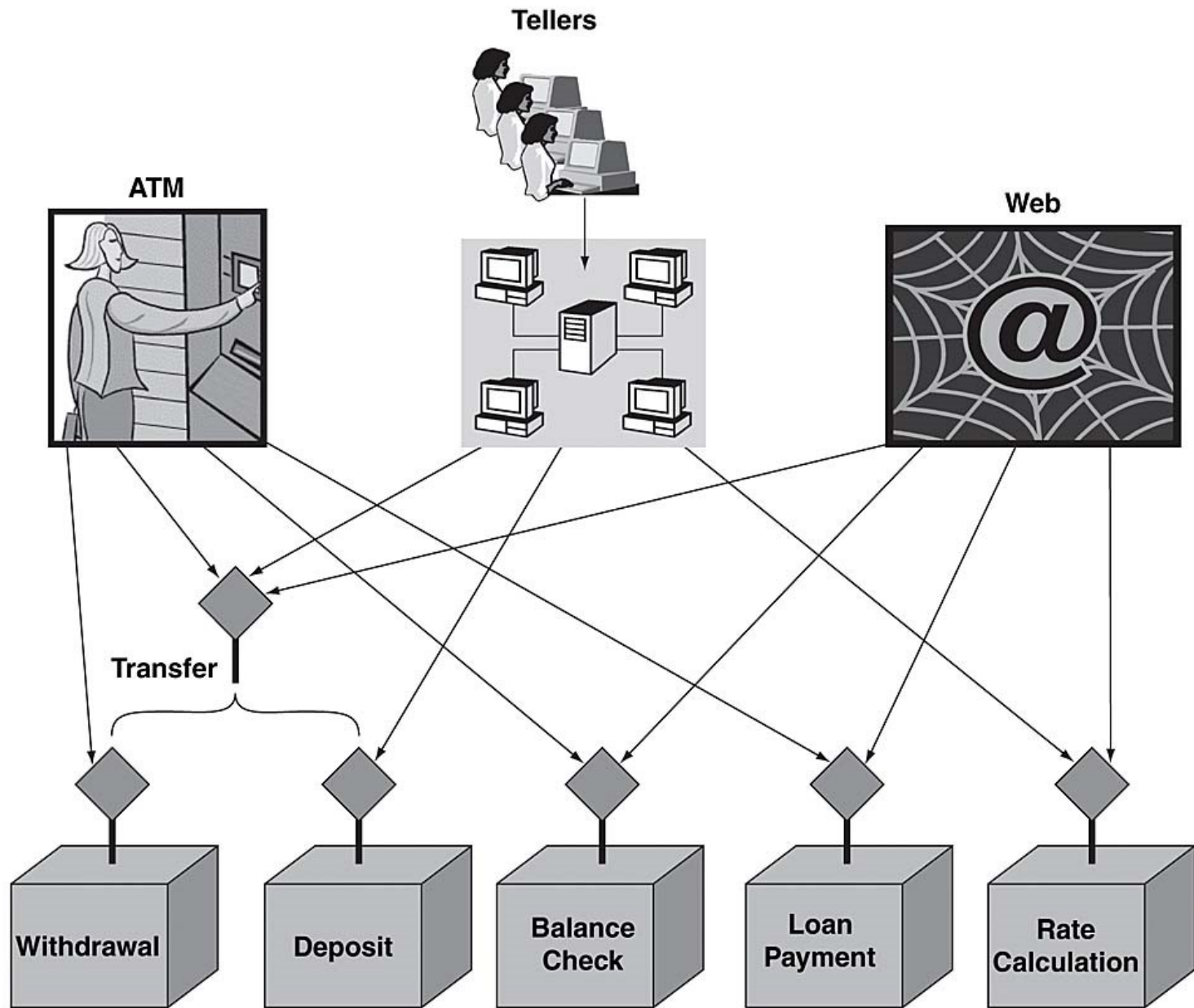
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What are Services?

- Most organizations (whether commercial or government) provide services to customers, clients, citizens, employees, or partners

Services

- **Bank** (as an Organization) provides services to its customers:
 - Account management (opening and closing accounts).
 - Loans (application processing, inquiries about terms and conditions, accepting payments)
 - Withdrawals, deposits, and transfers
 - Foreign currency exchange



Service-Oriented Architecture (wiki)

- **Service-oriented architecture (SOA)** is a style of software design where services are provided to the other components by application components, through a communication protocol over a network.
- A **SOA service** is a discrete unit of functionality that can be **accessed remotely** and **acted upon** and **updated independently**, such as retrieving a credit card statement online. SOA is also intended to be **independent of vendors, products** and **technologies**.

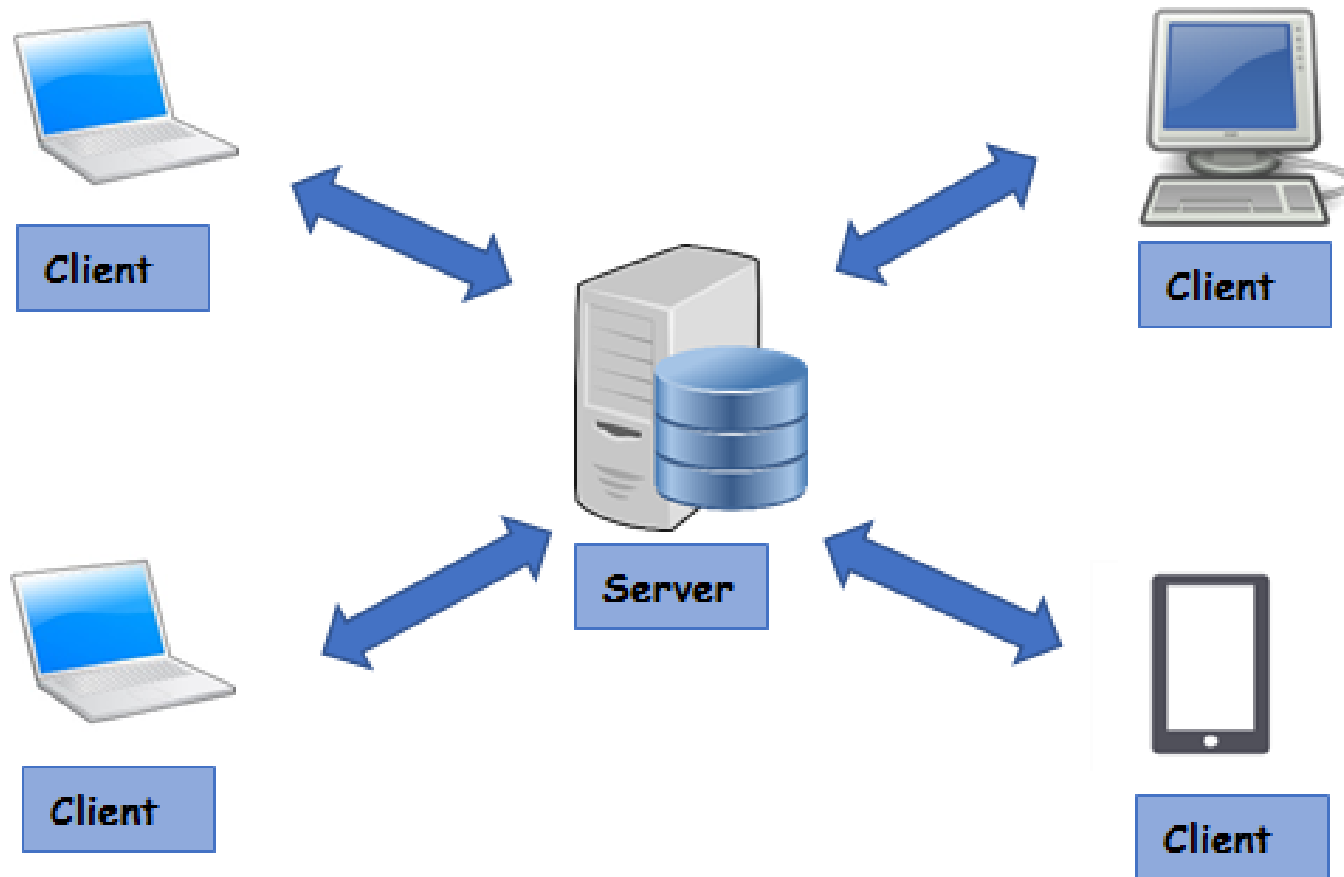
Service-Oriented Architecture

- Service-oriented architecture (SOA) is a **software development model** that allows services to **communicate** across different **platforms** and **languages** to form applications
- In SOA, a **service** is a **self-contained** unit of software designed to complete a specific task
- SOA allows various services to **communicate** using a **loose coupling** system to either pass data or coordinate an activity.

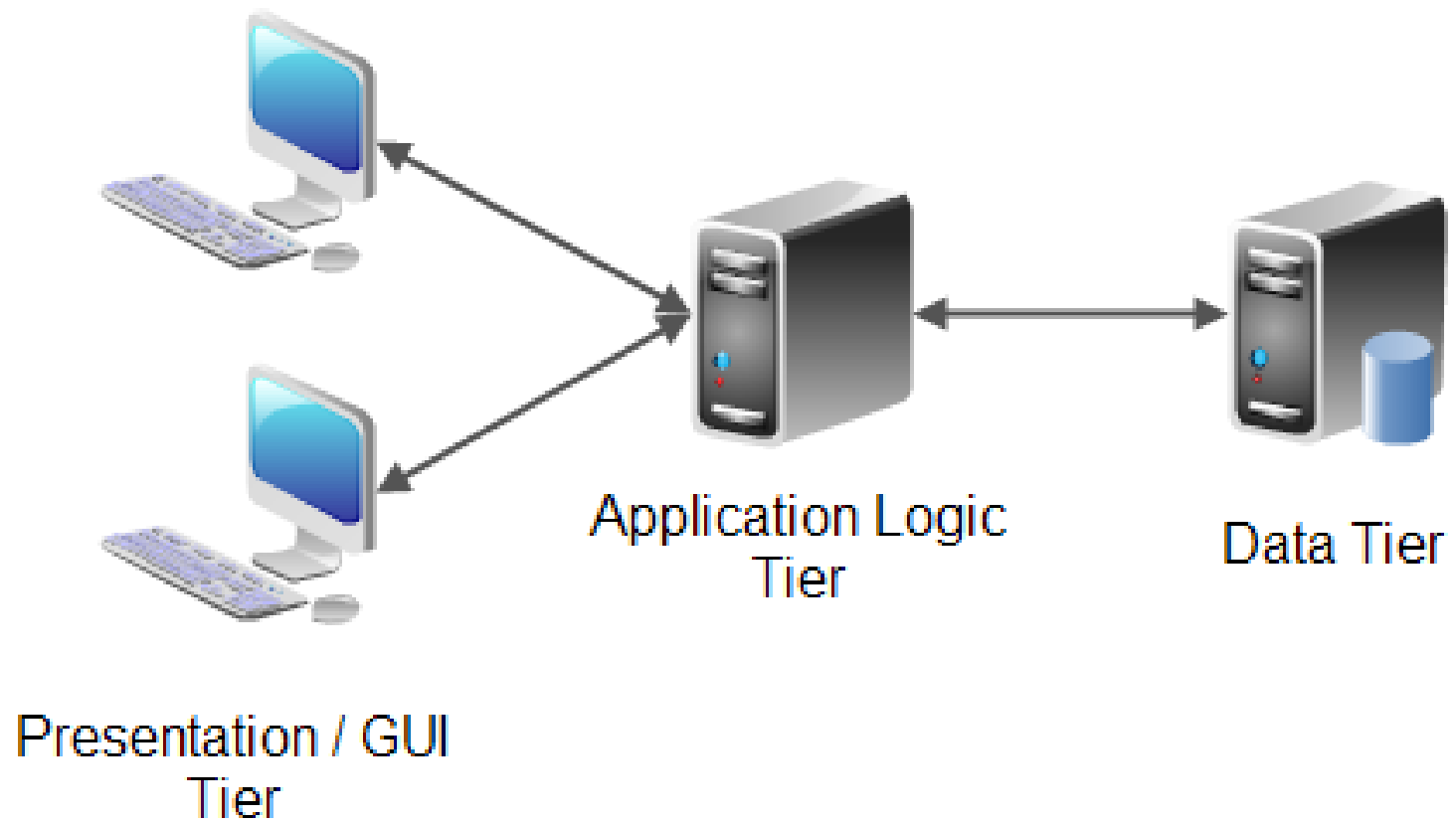
Evolution of SOA

- Procedural Programming
- Object Oriented Programming
- Centralized Systems
- Client Server Architecture
- N-tier Architecture
- RPC/RMI
- Distributed Architecture
- Cloud Computing

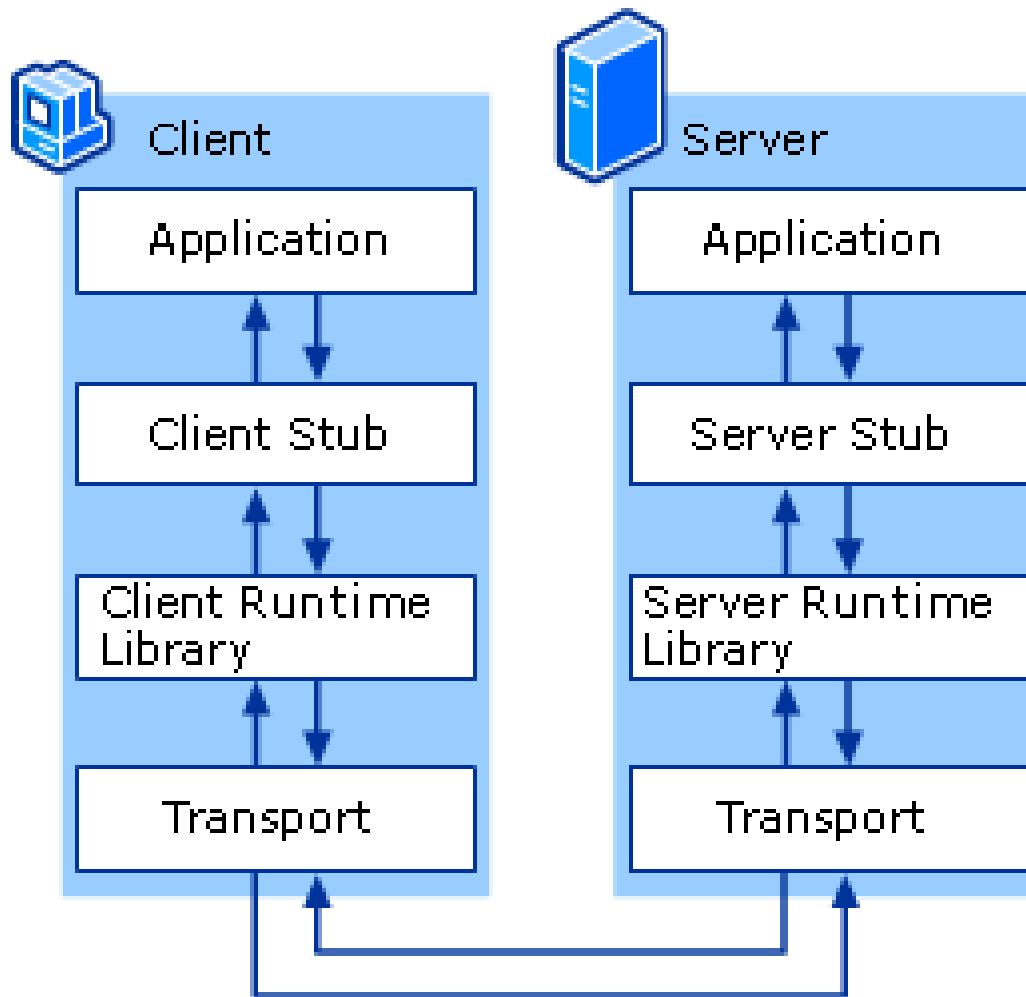
Client Server Architecture



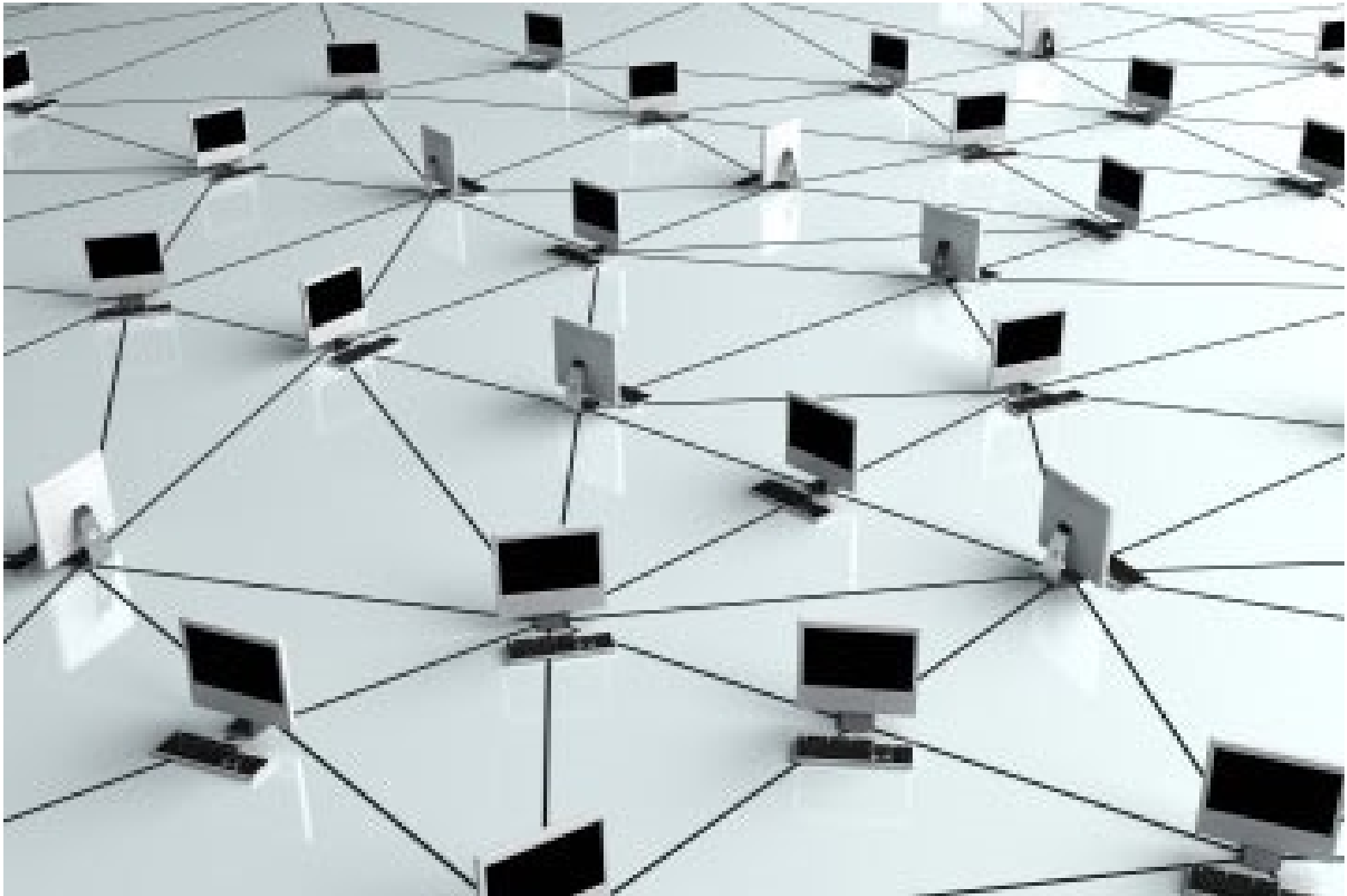
3-Tier Architecture



RPC Communication



Distributed Systems



Advantage of Distributed Systems

- Availability
- Fault tolerance and Recovery
- Concurrency
- Scalability
- Transparency
- Load balancing
- Performance
- Heterogeneity
- Openness
- Loose coupling

Web Service Roles

- **Service Provider**
 - The service provider implements the service and makes it available on the Internet.
- **Service Requestor**
 - This is any consumer of the web service
 - The requestor utilizes an existing web service by opening a network connection and sending an XML request.
- **Service Registry**
 - This is a logically centralized directory of services
 - The registry provides a central place where developers can publish new services or find existing ones

Web Service Protocol Stack

- **Service Transport**

- Responsible for transporting messages between applications
- HTTP, SMTP, Blocks Extensible Exchange Protocol (BEEP)

- **XML Messaging**

- Encoding messages in a common XML format so that messages can be understood at either end
- XML-RPC and SOAP

Web Service Protocol Stack

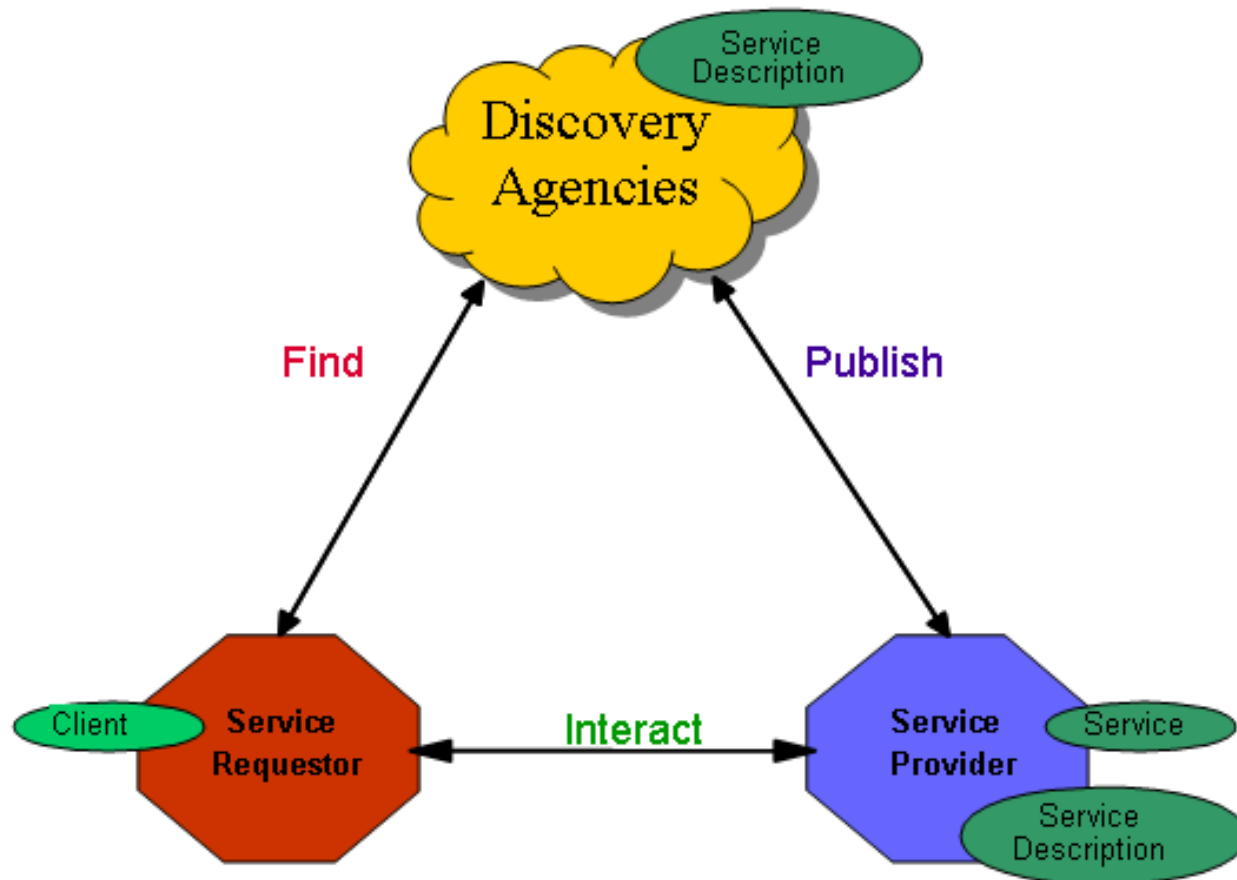
- **Service Description**

- describes the public interface to a specific web service
- Web Service Description Language (WSDL)

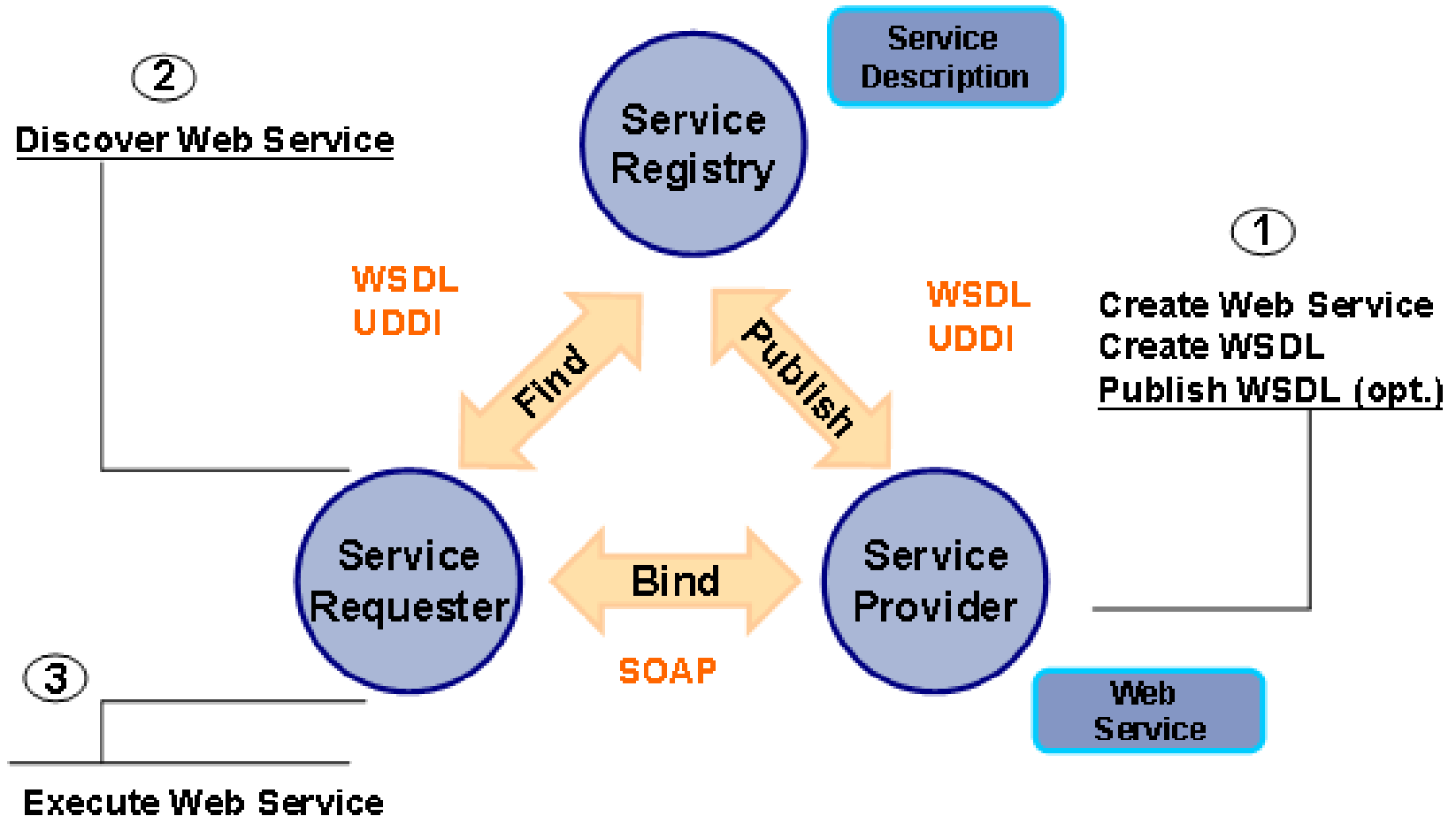
- **Service Discovery**

- centralizes services into a common registry and providing easy publish/find functionality
- Universal Description, Discovery, and Integration (UDDI)

Service Oriented Architecture



Service Oriented Architecture



Advantages of SOA

- Interoperability
- Usability
- Reusability
- Standardized Protocol
- Low cost communication
- Enterprise Application Integration

Web Services Examples

- Public/Private Web services
- <http://www.webservice.net/WS/wscatlist.aspx>
- Weather service
- Currency converter service
- Stock Market Quotes
- Login (google/facebook/linkedin)
- Payment (debit card/credit card/netbanking)
 - Support from different banks
- Advertizing (e.g. Amazon)

Web Services Components

- XML-RPC
- SOAP
- WSDL
- UDDI
- XML
- JSON

XML-RPC

- protocol for exchanging information
- uses XML messages to perform RPCs
- Requests are encoded in XML and sent via HTTP POST
- XML responses are embedded in the body of the HTTP response
- platform-independent
- allows diverse applications to communicate
 - e.g. A Java client can speak XML-RPC to a Perl server
- Easy to get started with web services

SOAP

- communication protocol
- format for sending messages
- designed to communicate via Internet
- platform independent
- language independent
- simple and extensible
- allows you to get around firewalls
- SOAP version 1.2 is W3C standard

WSDL

- stands for Web Services Description Language
- was developed jointly by Microsoft and IBM
- An XML based protocol for information exchange in decentralized and distributed environments
- WSDL definition describes how to access a web service and what operations it will perform
- is a language for describing how to interface with XML-based services
- is an integral part of UDDI, an XML-based worldwide business registry

Universal Description, Discovery, and Integration UDDI

- is an XML-based standard for
 - describing, publishing, and finding web services
- is a specification for a distributed registry of WS
- Is platform independent, open framework
- can communicate via SOAP, CORBA, and Java RMI Protocol
- uses WSDL to describe interfaces to web services
- is an open industry initiative enabling businesses to discover each other and define how they interact over the Internet

References

- Books:
 - “Service Oriented Architecture: Concepts, Technology and Design” by Thomas Erl, 1st Edition, Pearson publication
 - “Understanding SOA with Web Services” by Greg Lomow and Eric Newcomer, 1st Edition, Pearson Publication

References

- Web:
 - <https://www.tutorialspoint.com/webservices/>